

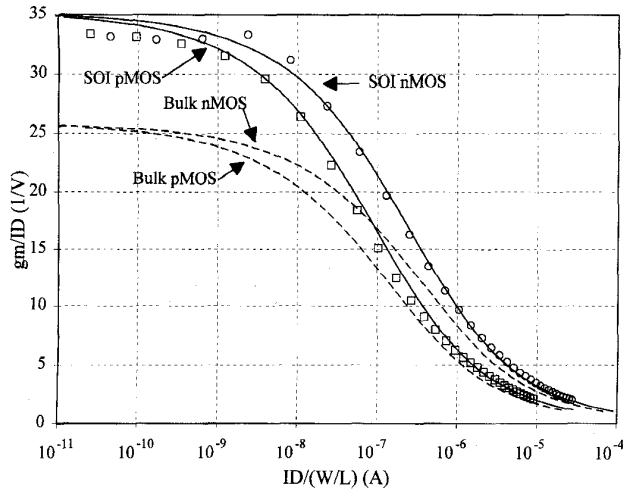


Fig. 1. Calculated (applying the model proposed in [7]) and measured g_m/I_D versus $I_D/(W/L)$ curves for n MOS and p MOS thin-film SOI fully-depleted transistors and n MOS and p MOS bulk transistors (Calculated SOI: solid line, measured SOI: circles (n MOS), squares (p MOS), calculated bulk: dashed line).

sometimes interpreted as a measure of the “transconductance generation efficiency” [6].

The relation of the g_m/I_D ratio with the transistor operating mode can be observed from the fact that this ratio is equal to the derivative of the logarithmic of I_D with respect to V_G as shown below

$$\frac{g_m}{I_D} = \frac{1}{I_D} \frac{\partial I_D}{\partial V_G} = \frac{\partial(\ln I_D)}{\partial V_G} = \frac{\partial \left\{ \ln \left[\frac{I_D}{\left(\frac{W}{L} \right)} \right] \right\}}{\partial V_G}. \quad (3)$$

This derivative is maximum in the weak inversion region where the I_D dependence versus V_G is exponential while it is quadratic in strong inversion, becoming almost linear deeply in strong inversion because of the velocity saturation. The maximum is equal to $1/(nU_T)$ where n is the subthreshold slope factor and U_T the thermal voltage. The g_m/I_D ratio decreases as the operating point moves toward strong inversion when I_D or V_G are increased as shown in Fig. 1. Therefore, the g_m/I_D ratio is also an indicator of the mode of operation of the transistor.

Let us now consider the dependence of g_m/I_D on transistor size. The normalized current $I_\square$ is independent of the transistors size. According to (3) the g_m/I_D ratio is also size independent. Therefore, the relationship between g_m/I_D and the normalized current is a unique characteristic for all transistors of the same type (n MOS or p MOS) in a given batch. Of course, this statement must be revised when dealing with short channel transistors.

The “universal” quality of the g_m/I_D versus $I_\square$ curve can be extensively exploited during the design phase, when the transistor aspect ratios (W/L) are unknown. Once a pair of values among g_m/I_D , g_m , and I_D has been derived, the W/L of the transistor can be determined unambiguously. An example of this procedure is described in the next section.

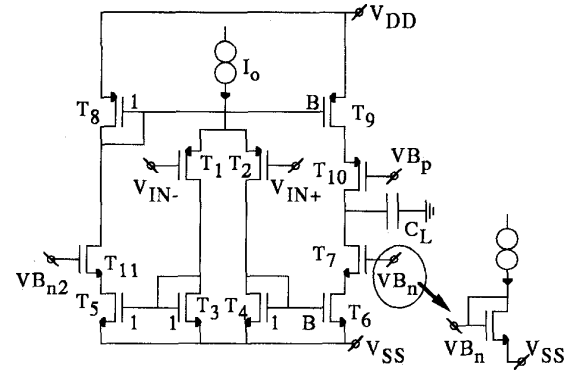


Fig. 2. SOI micropower OTA basic schematic.

The actual g_m/I_D versus $I_\square$ curve can be obtained in two ways: either analytically, using a MOS transistor model that provides a continuous representation of the transistor current and small-signal parameters in all regions of operations (like the model proposed in [7]), or from measurements on a typical transistor. It is more appropriate of course to consider a mean curve which is representative of a large number of transistors in order to take into account technology spreads [11]. We used the experimental approach for evaluation of the data that were used for the design of the SOI micropower OTA presented in the next section.

Fig. 1 shows the calculated and measured plots of g_m/I_D versus $I_\square$ for n and p MOS thin-film SOI fully-depleted transistors as well as for bulk transistors. The calculated plain curves were obtained applying the model presented in [7]. The bulk transistors curves were calculated assuming the same substrate concentration for the n and p MOS transistors. The different evolution of n and p MOS g_m/I_D curves with $I_\square$ is related to the mobility difference. As shown in Fig. 1, the increased subthreshold slope of the thin-film SOI transistors provides a maximum value of g_m/I_D of about 35 V^{-1} for the SOI transistors (which corresponds to $n = 1.1$) while only 25 V^{-1} is found for bulk transistors ($n = 1.5$) ([11]).

III. APPLICATION TO THE SYNTHESIS OF A SILICON-ON-INSULATOR MICROPPOWER CMOS OTA

The proposed methodology was applied to the synthesis of the cascoded OTA [12] shown in Fig. 2. We assume that the total supply current (I_{tot}) is known *a priori* and equal to $2 \mu\text{A}$. Furthermore, we assume that the load capacitor (C_L) is equal to 10 pF and the supply voltage (V_{DD}) equal to 3 V . The design procedure is illustrated here aiming at the best performances in terms of: dc open loop gain (A_0), transition frequency (f_T), phase margin (PM), and slew rate (SR). However, it can straightforwardly be modified to take into account other performance aspects (like noise or common mode rejection) that may be relevant to the particular application, as long as they are directly related to the unified g_m/I_D versus $I_\square$ relationship, i.e., to the current and small-signal parameters. For large-signal related performance aspects such as signal swing, an $I_D - V_G$ or $g_m/I_D - V_G$ relationship